

Improving Temporal Alignment and Feature Representation for Remote Sensing Video Super-Resolution

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Abstract—Remote sensing video super-resolution reconstructs high-resolution satellite video sequences from low-resolution observations. A central difficulty of this task is temporal alignment: satellite videos often contain platform-induced frame-level displacement, residual jitter, weak textures, and local structural variations, so useful cross-frame information can be blurred or misaggregated when alignment is inaccurate. This paper presents THGNet, a remote sensing VSR framework built on a recurrent backbone. Its key component is the Global-Local Offsets Estimator (GLOE), which refines second-order deformable alignment through complementary global-context recalibration and multi-scale local offset refinement. To support this alignment-centered design, an MAE-inspired temporal feature-masking pretraining strategy (T-MAE) provides a context-sensitive encoder initialization, and the High-Frequency Enhancement (HFE) module injects gated high-pass cues before temporal propagation. We also construct CQ1-VSR from raw unregistered satellite source videos to complement preprocessed benchmarks with stronger inter-frame variation. Under the reported comparison settings, THGNet achieves the best performance among the compared methods, reaching 39.68 dB PSNR on SAT-MAT-VSR and 41.40 dB on CQ1-VSR with 8.2M parameters. Code is available at <https://github.com/HIAS-Zhao/THGNet>.

Index Terms—Remote sensing video super-resolution, Masked feature pretraining, High-frequency enhancement, Deformable alignment.

I. Introduction

REMOTE sensing videos provide continuous observations of large-scale scenes and support applications such as geospatial monitoring, urban analysis, disaster response, and moving-target inspection. Compared with a single remote sensing image, a satellite video contains

complementary observations across time and therefore offers additional evidence for recovering roads, buildings, coastlines, and small objects. However, the spatial resolution of satellite videos is restricted by imaging distance, payload capacity, sensor bandwidth, and downlink transmission. Compression and downsampling further weaken fine textures and structural boundaries. Remote sensing video super-resolution (RSVSR), which reconstructs high-resolution (HR) frames from low-resolution (LR) video sequences, is therefore important for improving the usability of satellite observations [1]–[5].

The key advantage of video super-resolution (VSR) over single-image restoration is its ability to aggregate complementary information from neighboring frames [6]–[10]. To exploit this information, features belonging to the same scene content must first be aligned. This requirement is particularly challenging for satellite videos: platform motion, camera tremor, and residual registration errors can produce a scene-wide displacement, while parallax, moving objects, and fine structures still require spatially varying local correction. When these motion patterns coexist, inaccurate alignment can mix features from different structures, leading to blurred road intersections, softened building contours, and temporally unstable small objects. Reliable alignment under mixed scene-level and local motion is therefore a central challenge in RSVSR.

Considerable progress has been made in temporal propagation and alignment. Recurrent frameworks such as BasicVSR [11] and BasicVSR++ [12] propagate information over long sequences, while optical-flow and deformable-convolution methods refine cross-frame correspondence [13]–[17]. Recent RSVSR studies further explore efficient recurrent aggregation, temporal differences, feature diversity, deformable alignment, and lightweight bidirectional propagation [18]–[22]. Nevertheless, deformable offsets are commonly predicted through a unified convolutional mapping that must accommodate both scene-wide motion evidence and spatially varying corrections. Enlarging the receptive field supplies additional context, but does not explicitly provide complementary processing paths for these two roles. When repeated roofs, regular farmland grids, road-like textures, or weak boundaries provide insufficient correspondence cues, the predictor may sample from nearby but incorrect structures or fail to capture fine local variations. A natural question therefore arises: can offset estimation exploit scene-level context while retaining flexible refinement for spatially varying structures?

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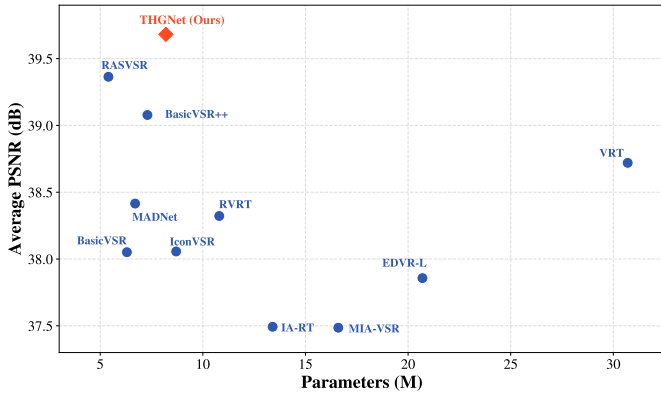


Fig. 1. Parameters (M) and PSNR (dB) performance comparison. Our THGNet obtains the highest PSNR among the compared VSR methods while using 8.2M parameters. Comparisons are conducted on the SAT-MAT-VSR val set.

To address this question, we propose THGNet, a recurrent framework for remote sensing video super-resolution. Its central component is the Global-Local Offsets Estimator (GLOE), which replaces the unified predictor in second-order deformable alignment with two complementary branches. The global branch uses channel-spatial recalibration to emphasize scene-level alignment evidence, while the local branch employs a dilated-convolution pyramid to extract local-to-medium-range cues for spatially varying refinement. Their outputs are combined to produce the final deformable offsets. In this manner, GLOE introduces both broad contextual guidance and local adaptive correction into offset estimation without changing the recurrent propagation pipeline.

Beyond alignment, the quality of the propagated representation also affects local reconstruction. Under uncertain or incomplete LR observations, distortion-oriented training may favor conservative averages, leaving roads, roof edges, and small objects over-smoothed even when PSNR or SSIM is high. We therefore introduce T-MAE, an MAE-inspired temporal feature-masking pretraining strategy that reconstructs the HR center frame from partially masked temporal features. This task provides the downstream encoder with a context-sensitive initialization for structural reconstruction. In addition, weak high-frequency signals may be attenuated during repeated feature extraction and recurrent propagation. HFE addresses this issue by injecting gated high-pass residual cues before temporal aggregation. T-MAE and HFE thus serve as supporting representation designs for contextual structure recovery and detail preservation, respectively.

We further construct CQ1-VSR from raw-source unregistered satellite videos to complement existing preprocessed benchmarks. The dataset retains stronger inter-frame jitter and local temporal variation, providing a more demanding setting for evaluating alignment under mixed motion. Experiments on SAT-MAT-VSR and CQ1-VSR show favorable reconstruction performance under the

reported settings. As shown in Fig. 1, THGNet obtains the highest PSNR among the compared methods with 8.2M parameters. The cumulative component study further indicates that GLOE provides the largest incremental gain, while T-MAE and HFE contribute additional improvements.

To sum up, the main contributions are summarized as follows:

- 1) We propose THGNet, an RSVSR framework whose core GLOE module combines global-context recalibration and multi-scale local refinement for deformable offset estimation.
- 2) We introduce two supporting feature designs: T-MAE uses temporal feature masking to learn context-sensitive representations, while HFE preserves weak high-frequency cues before recurrent propagation.
- 3) We construct CQ1-VSR from raw unregistered satellite videos and conduct extensive experiments on SAT-MAT-VSR and CQ1-VSR under stronger temporal variation.

The remainder of this paper is organized as follows. Section II reviews related work on natural and remote sensing video super-resolution. Section III presents the overall framework and the proposed GLOE, T-MAE, and HFE modules. Section IV reports the experimental results and ablation studies. Finally, Section V concludes this paper.

II. Related Work

A. Image and Natural Video Super-Resolution

Image super-resolution reconstructs HR images from LR observations. Early CNN-based methods such as SRCNN [23] and VDSR [24] demonstrated the effectiveness of end-to-end mapping, while deeper residual and attention-based networks such as EDSR [25] and RCAN [26] further improved spatial detail restoration. However, single-image methods rely only on intra-frame information and cannot exploit the temporal redundancy that naturally exists in video sequences.

Video super-resolution extends image super-resolution by aggregating information from adjacent frames. The key to VSR lies in how to explore spatial-temporal information while handling inter-frame misalignment. Existing VSR methods can be roughly grouped into sliding-window and recurrent frameworks. Sliding-window methods reconstruct a target frame from a limited temporal neighborhood and usually follow an alignment-fusion-reconstruction pipeline. Early and representative models include VSRnet [6], VESPCN [7], DUF [8], RBPN [10], and EDVR [15]. EDVR adopts pyramid, cascading, and deformable alignment to compensate neighboring features before fusion. Recent attention-based and transformer-based methods, including MuCAN [27], TGA [28], MANA [29], TTVSR [30], RVRT [31], VRT [32], MIA-VSR [33], and IA-RT [34], improve correspondence

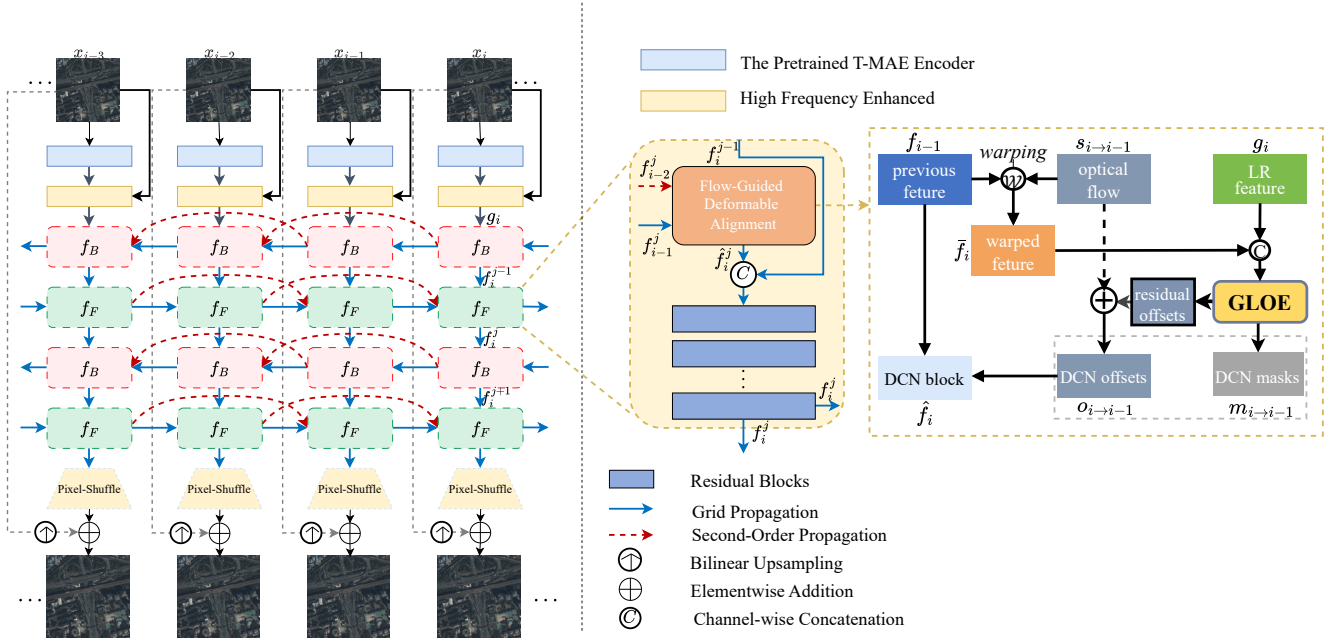


Fig. 2. Overall architecture of THGNet. Given an LR satellite video sequence, the model extracts frame-wise spatial features, supplements them with high-frequency cues, and performs recurrent propagation. The key alignment component, GLOE, refines second-order deformable alignment through global-context recalibration and multi-scale local refinement. The propagated forward/backward temporal features are finally fused and upsampled to reconstruct the HR video frames, while the feature extractor is initialized by T-MAE pretraining.

modeling, long-range dependency modeling, or implicit alignment.

Recurrent methods propagate features along the sequence and therefore make better use of information beyond a short local window. FRVSR [9], RSDN [35], BasicVSR [11], and BasicVSR++ [12] are representative recurrent frameworks, among which BasicVSR++ strengthens propagation with second-order alignment. These methods have shown strong performance on natural-video benchmarks and provide useful backbones for later RSVSR studies. However, natural videos usually contain richer object textures and more diverse foreground motion, whereas remote sensing videos often exhibit weak textures, repeated geospatial patterns, and platform-induced frame-level displacement. Directly transferring generic VSR designs may therefore leave the satellite-specific alignment ambiguity insufficiently addressed: a unified offset predictor must explain both the coarse displacement shared by most regions and the local residual corrections required by roads, buildings, and weak object boundaries.

B. Remote Sensing Video Super-Resolution

Remote sensing VSR has attracted increasing attention because satellite videos provide continuous observations for dynamic geospatial analysis, but their spatial resolution is constrained by payload, bandwidth, and transmission limitations. Early studies [1]–[3] adapted image restoration networks to satellite videos and showed that deep learning can improve video satellite imagery. These methods demonstrated the feasibility of satellite video

restoration, but they often relied heavily on image-level restoration or local temporal aggregation, leaving the spatial-temporal characteristics of satellite videos insufficiently explored.

Later methods began to explicitly consider temporal redundancy and remote-sensing-specific properties. Liu et al. [4] exploited adaptively selected spatial-temporal neighbors and nonlocal similarity for alignment, and further investigated satellite VSR under multiple degradations [36]. He and He [37] proposed a unified network for arbitrary-scale satellite video super-resolution, and Chen et al. [38] studied robust extraction and super-resolution of low-resolution flying airplanes from satellite video. MSDTGP [5] introduced multi-scale deformable convolution alignment and temporal grouping projection, while LGTD [19] used local-global temporal difference learning to improve temporal modeling.

More recent RSVSR works further explore recurrent propagation, feature representation, and efficient modeling. RASVSR [18] designs a lightweight recurrent aggregation framework with second-order alignment for long-range satellite video sequences. MADNet [20] improves spatial-temporal representation by enhancing feature diversity in spatial and frequency dimensions, indicating that satellite videos can benefit from more diverse feature encoding. EBRNet [22] develops an enhanced bidirectional recurrent network with multi-branch frequency-spatial enhancement and multiscale second-order alignment, emphasizing long-range aggregation, lightweight design, and scale-diverse alignment. Collectively, these studies moti-

vate architectures adapted to the temporal and spatial characteristics of satellite videos.

Nevertheless, several issues remain insufficiently explored. First, temporal alignment is still commonly handled by a unified offset estimation process. When raw or weakly registered satellite videos contain both large frame-level shifts and fine local variations, a single mapping may have difficulty balancing scene-level context and local correction, especially in repeated or weak-texture regions. Second, metric-oriented reconstruction may favor conservative local structures: a result can obtain high PSNR/SSIM while road intersections, building contours, or small objects remain over-smoothed. A pretraining task that encourages context-sensitive representation for remote sensing video features is therefore worth exploring. Third, high-frequency cues are often embedded implicitly in feature blocks. In satellite videos, weak edges, boundaries, and fine spatial details may be attenuated during recurrent propagation, making them difficult to recover at the final reconstruction stage.

C. Temporal Alignment and Feature Representation

Accurate temporal alignment is a fundamental requirement in VSR. Sliding-window methods usually follow an alignment-fusion-reconstruction pipeline, where neighboring frames are first compensated toward the target frame and then aggregated for reconstruction. Explicit alignment methods estimate motion fields and use optical-flow warping for frame- or feature-level compensation. Representative works use pre-trained flow estimators such as SPyNet [39], learn task-oriented flows for video enhancement [13], or estimate HR optical flow to provide more accurate compensation cues [40]. VESPCN [7] also combines spatial-temporal networks with motion compensation for real-time VSR.

Implicit alignment methods avoid relying solely on a fixed optical-flow field and instead embed alignment into learnable spatial-temporal operators. Early CNN-based VSR models aggregate adjacent frames with convolutional networks [6], while DUF [8] learns dynamic upsampling filters for implicit motion handling. 3D convolution and spatial-temporal filtering have also been used to jointly encode local spatial-temporal context, including satellite VSR settings with multi-scale 3D convolution [37]. Deformable convolution further improves adaptive alignment by learning sampling offsets [16], [17]. TDAN [14] predicts deformable offsets for temporal alignment, EDVR [15] adopts pyramid, cascading, and deformable alignment, and BasicVSR++ [12] strengthens recurrent propagation with second-order deformable alignment. In RSVSR, MS-DTGP [5] and recent continuous-scale satellite VSR methods [21] also demonstrate the usefulness of deformable alignment for satellite videos.

These alignment studies have greatly improved spatial-temporal aggregation, but satellite videos can contain both large frame-level displacement and fine-grained local variations. We do not claim that previous predictors have

no access to global context; rather, their offset estimation is usually optimized as one unified mapping. A unified predictor may become unreliable when platform jitter produces coherent scene-wide motion while different local structures still require distinct corrections. Introducing complementary global-context and local-refinement paths is therefore a natural extension for RSVSR alignment.

Beyond temporal alignment, representation learning also plays an important role in visual restoration and video understanding. Masked autoencoders [41], [42] learn representations by reconstructing masked content from visible spatial or temporal context. Such reconstruction-oriented tasks are relevant to VSR because high-resolution details are not directly observed in LR inputs and must be estimated from incomplete evidence across frames. Most existing masked modeling studies, however, are designed for generic image or video representation learning. Our T-MAE is inspired by this principle but applies masking at the encoded-feature level and reconstructs an HR center frame from a masked temporal representation, adapting the pretraining task to the downstream RSVSR pipeline.

High-frequency representations are particularly important for restoration because edges, boundaries, and fine textures strongly affect perceptual and reconstruction quality [43]–[45]. These studies suggest that high-frequency cues can provide useful complementary information for recovering local details. In recurrent VSR, however, weak detail cues may be repeatedly aligned and aggregated before the reconstruction head. This motivates a lightweight design that supplements backbone features with high-frequency residual information after feature extraction and before recurrent propagation, rather than treating high-frequency modeling as a separate heavy branch.

III. Method

A. Overview

The overall architecture of THGNet is shown in Fig. 2. Given an LR video sequence $\mathbf{X} = \{I_1, \dots, I_T\}$, THGNet reconstructs the HR sequence $\hat{\mathbf{Y}} = \{\hat{Y}_1, \dots, \hat{Y}_T\}$ with an upscaling factor of 4. The model follows the recurrent propagation structure of BasicVSR++ and keeps the main alignment-fusion-reconstruction pipeline unchanged. Our main modification is made at the second-order deformable alignment stage, where GLOE combines global-context recalibration with multi-scale local offset refinement.

More specifically, the feature extractor first maps each LR frame into a spatial feature representation. HFE supplements this representation with gated high-pass residual cues before temporal propagation. The enhanced features are then propagated in forward and backward directions. During propagation, optical flow provides a coarse motion prior, while deformable alignment refines feature sampling. GLOE improves this stage by combining scene-level contextual evidence with multi-scale local cues for offset prediction. Finally, the spatial feature of the current frame and the propagated temporal features from

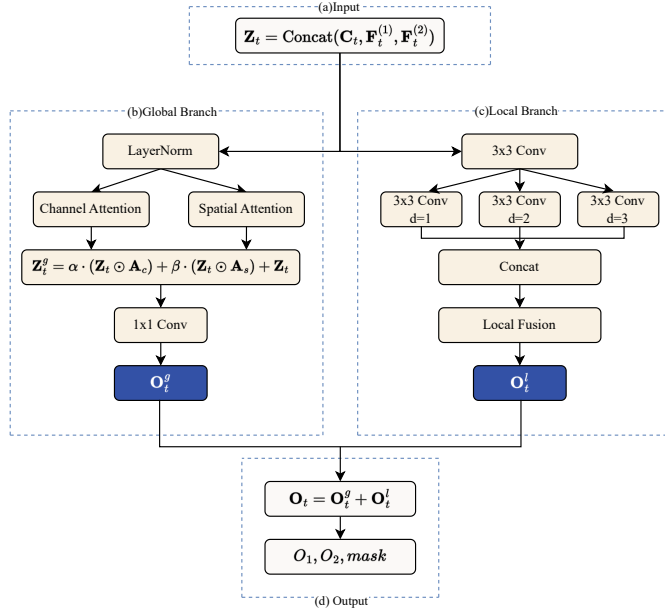


Fig. 3. Structure of the GLOE module for offset prediction. The global branch predicts global-context-aware offsets, while the local branch predicts multi-scale local offsets through a dilated-convolution pyramid. The two branch outputs are added to obtain the final offsets for deformable alignment.

multiple branches are fused and upsampled by pixel-shuffle layers to produce the HR output.

This design treats temporal alignment as the main difficulty of RSVSR. T-MAE and HFE are introduced as supporting components: T-MAE provides a context-sensitive initialization for the feature extractor, and HFE preserves weak high-frequency cues that may otherwise be attenuated during recurrent propagation.

B. Global-Local Offsets Estimator

Building on the alignment motivation discussed in the Introduction, GLOE replaces the unified offset predictor in BasicVSR++ with a dual-branch estimator, as shown in Fig. 3. The recurrent propagation and deformable alignment pipeline remain unchanged, so GLOE focuses on the offset estimation stage. The global branch recalibrates the predictor input with lightweight channel-spatial attention to emphasize scene-level alignment evidence. The local branch complements it with a dilated-convolution pyramid, which extracts local-to-medium receptive-field cues for spatially varying refinement. The two branches are not constrained to form a unique physical decomposition of motion; instead, they provide complementary contextual representations for offset prediction while avoiding expensive all-pairs matching or heavy transformer blocks.

Let $\mathbf{Z}_t$ be the predictor input formed by concatenating conditional alignment features and first-/second-order optical-flow priors:

$$\mathbf{Z}_t = \text{Concat}(\mathbf{C}_t, \mathbf{F}_t^{(1)}, \mathbf{F}_t^{(2)}). \quad (1)$$

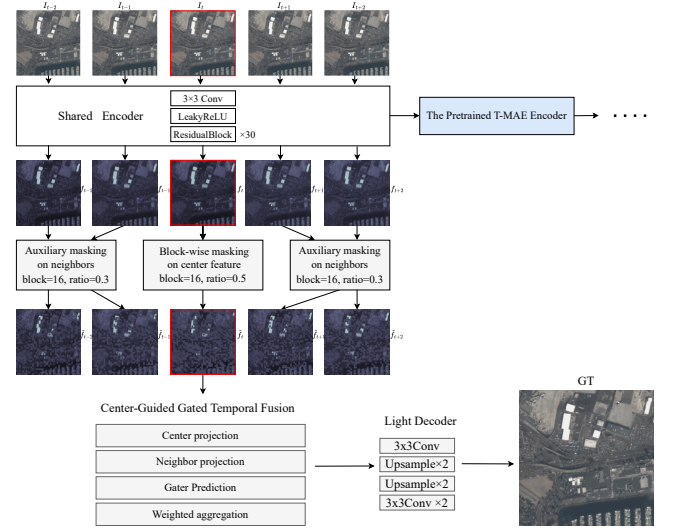


Fig. 4. Overview of T-MAE pretraining. For a five-frame LR input sequence, the center-frame feature is strongly block-masked while neighboring-frame features are lightly masked. A center-guided gated fusion module aggregates the masked temporal representation, and a lightweight decoder reconstructs the HR center frame. After pretraining, only the encoder is transferred to THGNet.

Here, $\mathbf{C}_t$ contains the feature conditions used for deformable alignment, and $\mathbf{F}_t^{(1)}$ and $\mathbf{F}_t^{(2)}$ provide coarse motion priors from first- and second-order propagation. Instead of directly feeding this concatenated tensor into a single predictor, the global branch first normalizes and recalibrates it with channel and spatial attention:

$$\bar{\mathbf{Z}}_t = \text{LN}(\mathbf{Z}_t), \quad (2)$$

$$\mathbf{Z}_t^g = \alpha(\bar{\mathbf{Z}}_t \odot \mathbf{A}_c) + \beta(\bar{\mathbf{Z}}_t \odot \mathbf{A}_s) + \bar{\mathbf{Z}}_t, \quad (3)$$

where $\text{LN}(\cdot)$ denotes layer normalization, $\mathbf{A}_c$ and $\mathbf{A}_s$ are channel and spatial attention maps, and α, β are learnable parameters. Specifically, $\mathbf{A}_c$ is generated by global average pooling followed by two 1×1 convolutions with a channel reduction ratio of four, a ReLU activation, and a sigmoid activation. To obtain $\mathbf{A}_s$, channel-wise average and maximum responses are concatenated and passed through a 7×7 convolution followed by a sigmoid activation. The residual connection keeps the original flow-conditioned information available, while the attention terms emphasize channels and regions that may be informative for scene-level alignment. A lightweight prediction head then maps the recalibrated representation to the global-branch offsets:

$$\mathbf{O}_t^g = \mathcal{P}_g(\mathbf{Z}_t^g). \quad (4)$$

The local branch first compresses the predictor input and then adopts a dilated convolution pyramid:

$$\mathbf{Z}_t^r = \sigma(\text{Conv}_{3 \times 3}(\mathbf{Z}_t)), \quad (5)$$

$$\mathbf{Z}_t^l = \text{Concat}(\phi_{d_1}(\mathbf{Z}_t^r), \phi_{d_2}(\mathbf{Z}_t^r), \phi_{d_3}(\mathbf{Z}_t^r)). \quad (6)$$

Here, $\mathbf{Z}_t^r$ is the reduced local feature, $\mathbf{Z}_t^l$ denotes the concatenated multi-dilation local representation, $\phi_{d_i}(\cdot)$ denotes a 3×3 convolution with dilation rate d_i , and $\sigma(\cdot)$ is ReLU. Different dilation rates provide adjacent receptive-field scales without increasing the convolution kernel size. This design gives the local branch access to both small-neighborhood and medium-range contextual cues. The local fusion module then aggregates this representation and predicts the local-branch offsets:

$$\mathbf{O}_t^l = \mathcal{P}_l(\mathbf{Z}_t^l). \quad (7)$$

where $\mathcal{P}_l(\cdot)$ denotes the local fusion module composed of a 1×1 convolution, ReLU, and a final 1×1 prediction layer.

The final prediction is obtained by fusing the two branch outputs:

$$\mathbf{O}_t = \mathbf{O}_t^g + \mathbf{O}_t^l. \quad (8)$$

The local branch uses the dilation configuration (1, 2, 3) by default. This choice provides continuous local-to-medium receptive-field scales for local motion modeling while avoiding overly sparse sampling. We further evaluate (1, 1, 1), (1, 2, 4), and (1, 3, 5) in Sec. IV-D to examine whether larger or sparser dilation improves offset prediction.

C. Temporal Masked Pretraining

To improve local structural recovery in weak-texture remote sensing videos, we introduce an MAE-inspired temporal feature-masking pretraining strategy, termed T-MAE, as shown in Fig. 4. The purpose of T-MAE is to provide a context-sensitive initialization for the downstream encoder. Existing RSISR models are commonly optimized for pixel-wise reconstruction metrics such as PSNR and SSIM. These metrics are important, but optimization under uncertain LR evidence may still produce conservative or over-smoothed local structures. In satellite videos, weak road lines, building contours, and small object boundaries may be difficult to identify from the LR center frame alone, while neighboring frames provide complementary observations. T-MAE therefore reconstructs the HR center frame from a partially masked temporal feature representation, encouraging the pretraining pipeline to use spatial and temporal context.

Given a five-frame LR input sequence centered at time t ,

$$\mathbf{X}_t = \{I_{t-2}, I_{t-1}, I_t, I_{t+1}, I_{t+2}\}, \quad (9)$$

a shared encoder $\mathcal{E}(\cdot)$ extracts frame-wise features:

$$f_\tau = \mathcal{E}(I_\tau), \quad \tau \in \{t-2, t-1, t, t+1, t+2\}. \quad (10)$$

First, we apply block-wise masking to the center-frame feature f_t . Let the spatial size of f_t be $H_f \times W_f$. We partition it into non-overlapping blocks of size $P \times P$ and randomly mask a proportion ρ_c of the blocks. The masked center feature is

$$\tilde{f}_t = f_t \odot (1 - M_t), \quad (11)$$

where M_t is the binary block mask. Because masking is applied after frame-wise encoding, T-MAE differs from conventional input-token MAE. The structured feature masking reduces the information directly available to the fusion module and decoder, encouraging greater use of the remaining spatial and temporal context.

Second, we apply auxiliary masking to neighboring-frame features. If neighboring features remain fully available, the fusion module may rely excessively on a small number of adjacent observations. Therefore, each neighboring feature f_k is also masked:

$$\tilde{f}_k = f_k \odot (1 - M_k), \quad k \in \{t-2, t-1, t+1, t+2\}, \quad (12)$$

where the neighboring masking ratio ρ_n is lower than ρ_c . This preserves useful temporal context while limiting direct reliance on fully visible neighboring features.

The masked temporal features are fused by center-guided gated fusion:

$$g_k = \sigma\left(\psi([\phi_c(\tilde{f}_t), \phi_n(\tilde{f}_k)])\right), \quad (13)$$

$$a_t = \phi_c(\tilde{f}_t) + \sum_{k \neq t} g_k \odot \phi_n(\tilde{f}_k), \quad (14)$$

$$h_t = \omega([\phi_c(\tilde{f}_t), a_t]), \quad (15)$$

where $\phi_c(\cdot)$ and $\phi_n(\cdot)$ are projection functions, $\psi(\cdot)$ predicts the gate, and $\omega(\cdot)$ denotes the final fusion module composed of two 3×3 convolutions, each followed by a LeakyReLU activation. A lightweight decoder reconstructs the HR center frame:

$$\hat{Y}_t = \mathcal{D}(h_t). \quad (16)$$

The pretraining objective combines a reconstruction term and a gradient term:

$$\mathcal{L}_{\text{T-MAE}} = \mathcal{L}_{\text{rec}} + 0.1\mathcal{L}_{\text{grad}}. \quad (17)$$

Here, $\mathcal{L}_{\text{rec}}$ is the Charbonnier reconstruction loss evaluated on the masked regions of the HR center frame, and $\mathcal{L}_{\text{grad}}$ is the gradient loss between the reconstructed and ground-truth HR center frames based on their Sobel gradient magnitudes. After pretraining, the decoder is discarded. The encoder is used to initialize the downstream THGNet feature extractor and is fine-tuned jointly with the VSR network unless otherwise specified. We also report a frozen-encoder setting in the ablation study to examine the effect of encoder updating.

D. High-Frequency Enhancement Module

High-frequency details such as edges, boundaries, and fine textures are important for structural reconstruction in remote sensing videos. Although deep VSR backbones can learn high-frequency information implicitly, repeated alignment and aggregation may attenuate weak high-frequency cues. HFE is therefore designed as a lightweight auxiliary branch that supplies high-pass residual information before temporal aggregation, allowing these cues to participate in propagation rather than being introduced

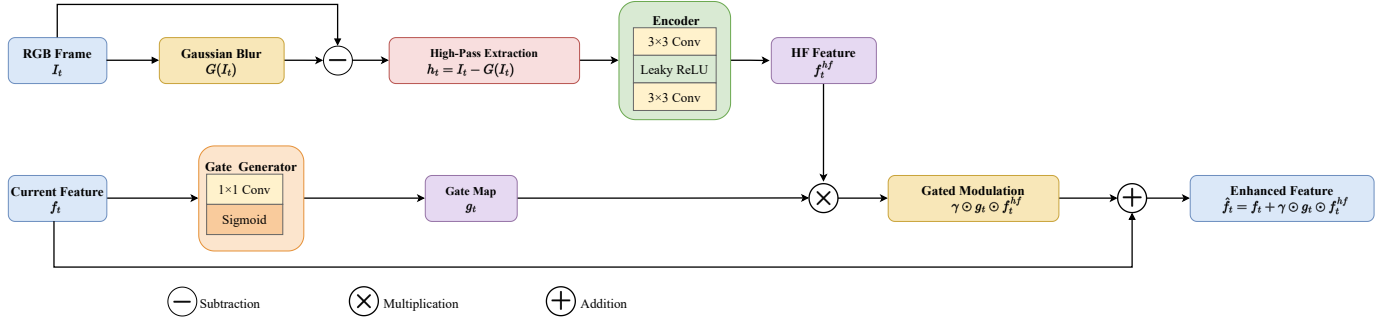


Fig. 5. Structure of the HFE module. The LR input is first decomposed by a high-pass operator to obtain image-level residual cues such as edges, boundaries, and fine textures. These cues are projected into the feature space and modulated by a feature-dependent gate before being added to the backbone feature, allowing weak high-frequency details to participate in temporal propagation rather than being recovered only at the reconstruction head.

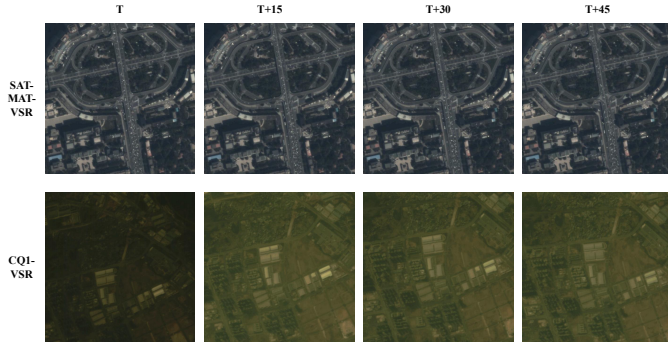


Fig. 6. Visual comparison between SAT-MAT-VSR and CQ1-VSR. The top row shows SAT-MAT-VSR samples with relatively stable object positions, while the bottom row shows CQ1-VSR samples from raw unregistered videos with visible platform motion and larger inter-frame displacement.

only at the reconstruction head. The structure of HFE is shown in Fig. 5.

Given an input frame I_t and its feature f_t , the default high-frequency residual is obtained by Gaussian high-pass filtering:

$$h_t = I_t - G(I_t), \quad (18)$$

where $G(\cdot)$ denotes Gaussian blur. The residual is projected to the feature space:

$$f_t^{hf} = \eta(h_t), \quad (19)$$

where $\eta(\cdot)$ is a small convolutional mapping. A gate g_t is predicted from f_t to control the injection strength:

$$\hat{f}_t = f_t + \gamma \cdot g_t \odot f_t^{hf}, \quad (20)$$

where γ is a learnable scalar. The gate allows the network to adjust the contribution of high-frequency residuals according to the backbone feature. To examine whether the observed gain depends on the default Gaussian residual, we additionally evaluate Laplacian, Sobel-gradient, and learned depth-wise filters in Sec. IV-D.

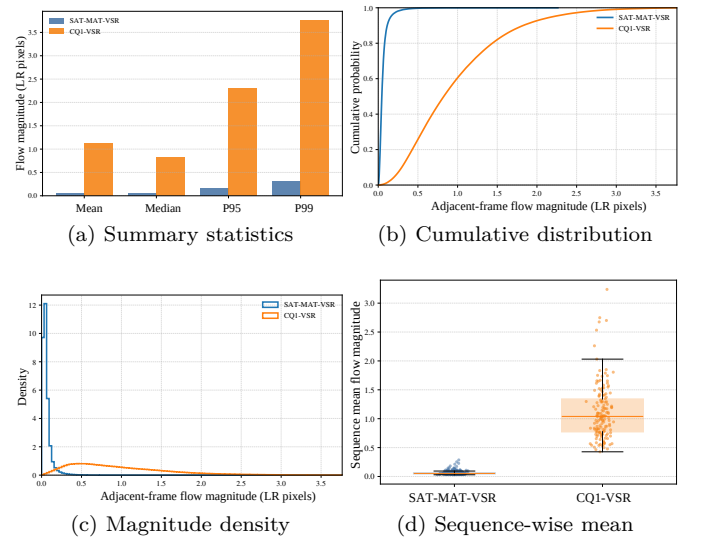


Fig. 7. Adjacent-frame optical-flow magnitude comparison between SAT-MAT-VSR and CQ1-VSR. The four statistics compare motion intensity from complementary perspectives, showing that stronger temporal variation in CQ1-VSR increases the difficulty of feature alignment in remote sensing video super-resolution.

IV. Experiments

A. Datasets and Evaluation Metrics

1) Datasets: Experiments are conducted on SAT-MAT-VSR [18] and CQ1-VSR. SAT-MAT-VSR is built from Jilin-1 satellite videos and contains 431 clips, including 413 training clips and 18 validation clips. Each clip contains 100 frames with HR resolution 640×640 , and LR frames are generated by $\times 4$ bicubic downsampling. This dataset serves as the main benchmark for comparison with existing RSVSR methods.

CQ1-VSR is constructed from raw-source remote sensing video sequences captured by the QK-1 satellite. Since the original data are stored in the Bayer pattern format, we first convert them into RGB images. The LR inputs are then generated with a clip-consistent $\times 4$ synthetic degradation pipeline adapted from Real-ESRGAN [47], where blur degradation and noise corruption are applied during

TABLE I

Quantitative Comparison in Terms of PSNR, SSIM, and NIQE. The Best and Second-Best Performances Are Highlighted in Red and Blue, Respectively.

Method	Frames	Params (M)	FLOPs (G)	Publication	SAT-MAT-VSR			CQ1-VSR		
					PSNR (dB)↑	SSIM↑	NIQE↓	PSNR (dB)↑	SSIM↑	NIQE↓
Bicubic	-	-	-	-	35.6158	0.9279	7.7071	37.1798	0.9305	14.8178
EDVR-L [15]	5	20.7	143.5	CVPRW 2019	37.8572	0.9537	7.2520	40.5867	0.9579	10.3482
BasicVSR [11]	15	6.3	396.9	CVPR 2021	38.0512	0.9551	7.0710	41.0533	0.9604	10.1047
IconVSR [11]	15	8.7	498.5	CVPR 2021	38.0566	0.9532	7.0704	40.7682	0.9585	9.9664
BasicVSR++ [12]	30	7.3	879.9	CVPR 2022	39.0868	0.9628	6.7360	41.2892	0.9610	9.9944
RVRT [31]	16	10.8	697.9	NeurIPS 2022	38.3221	0.9562	7.1867	41.0233	0.9604	10.1183
VRT [32]	16	30.7	1298.4	TIP 2024	38.7200	0.9588	7.0677	40.5266	0.9576	10.2260
RASVSR [18]	30	5.4	638.3	J-STARS 2024	39.3644	0.9643	6.8987	41.1596	0.9617	10.1140
MIA-VSR [33]	16	16.6	1376.9	CVPR 2024	37.4859	0.9490	7.4653	41.2675	0.9619	9.9192
IA-RT [34]	16	13.4	2228.1	CVPR 2024	37.4930	0.9492	7.2552	40.9804	0.9609	10.1864
MADNet [20]	15	5.8	318.6	TIP 2025	38.4151	0.9575	6.8104	41.0613	0.9609	10.0724
NSBRNet [46]	30	5.87	-	TCSVT 2026	38.75	0.9599	-	-	-	-
THGNet (Ours)	30	8.2	982.6	-	39.6817	0.9681	6.6974	41.4010	0.9625	9.8922

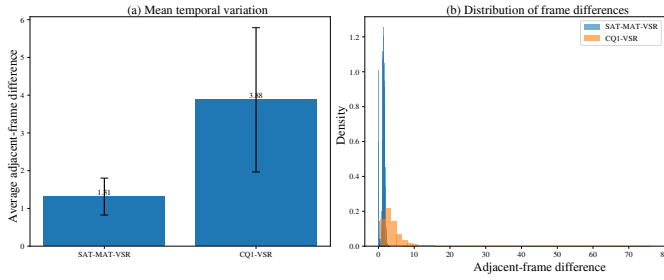


Fig. 8. Temporal variation comparison between SAT-MAT-VSR and CQ1-VSR. Adjacent-frame variation is computed on LR video frames to quantify how much the content changes from one frame to the next. Compared with SAT-MAT-VSR, CQ1-VSR shows stronger and more widely distributed temporal variation, providing a more demanding setting for temporal alignment modules.

preprocessing. CQ1-VSR should therefore be interpreted as a raw-source dataset with synthetically generated LR-HR pairs, rather than a dataset with physically captured paired LR and HR videos. After preprocessing, CQ1-VSR is split into 150 training sequences, 26 validation sequences, and 22 testing sequences. Each HR frame is cropped to 640×640 , and the corresponding LR frame has a spatial size of 160×160 . The dataset contains diverse scene categories, including urban areas, airports, coastal regions, and farmlands.

The two datasets play complementary roles. SAT-MAT-VSR measures performance on a standard RSVSR benchmark, while CQ1-VSR evaluates reconstruction on raw-source satellite scenes with broader scene distributions and stronger temporal variation. As shown in Fig. 6, Fig. 7, and Fig. 8, CQ1-VSR contains more visible inter-frame displacement and stronger adjacent-frame changes, providing a more demanding setting for temporal aggregation and alignment.

2) Evaluation Metrics: To evaluate the quality of reconstructed RSVSR videos, we use Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) [48] as full-reference evaluation metrics. Higher PSNR and SSIM values indicate better reconstruction quality. In addition,

the natural image quality evaluator (NIQE) [49] is adopted as a complementary no-reference quality indicator.

B. Implementation Details

1) Model Details: THGNet is implemented for $\times 4$ RSVSR. The feature channel number is 64. A pretrained SPyNet [39] is used to estimate optical flow, and its parameters are frozen for the first 5,000 training iterations. Each forward and backward propagation branch contains seven residual blocks. In T-MAE pretraining, five LR frames are used as input and the HR center frame is used as the reconstruction target. The encoder has 64 channels and 30 residual blocks. The center-frame masking ratio is 0.5 and the neighboring-frame masking ratio is 0.3.

2) Training Details: For downstream VSR training, unless otherwise specified, the feature extractor starts from the pretrained T-MAE encoder and is fine-tuned jointly with the full THGNet. We use Adam with $\beta_1 = 0.9$ and $\beta_2 = 0.99$, an initial learning rate of 1×10^{-4} , cosine annealing, 30 input frames, and Charbonnier loss. The LR patch size is 64×64 . Random flipping and rotation are used for augmentation. Training is conducted for 100,000 iterations. To reflect deployment cost, we report both parameter counts and FLOPs for each method.

For the random-initialization ablation, the same encoder architecture as T-MAE is initialized randomly and trained end-to-end with the downstream VSR objective. Our model is implemented in PyTorch and trained on two NVIDIA H200 GPUs with 141 GB memory, along with an Intel Xeon Platinum 8558 CPU at 2.1 GHz. The full training process takes approximately three days.

C. Comparison With Representative Methods

To evaluate the performance of THGNet, we compare it with bicubic interpolation and representative approaches, including the sliding-window method EDVR-L [15], recurrent VSR methods BasicVSR [11], IconVSR [11], BasicVSR++ [12], and RVRT [31], transformer-based VSR methods VRT [32], MIA-VSR [33], and IA-RT [34], and recent remote sensing VSR methods RASVSR [18],

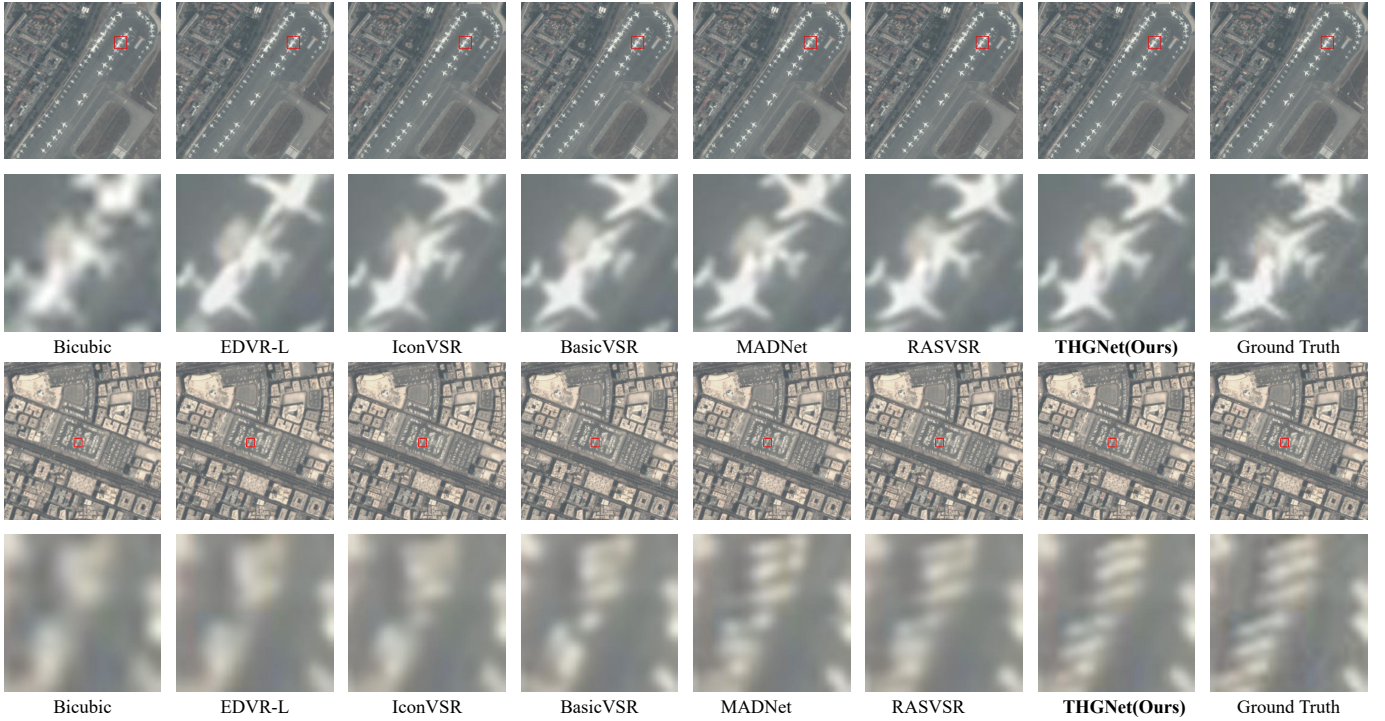


Fig. 9. Qualitative comparison on SAT-MAT-VSR. The scene at the top belongs to val clip 008 and the scene below belongs to val clip 009. Zoomed in for better visualization.

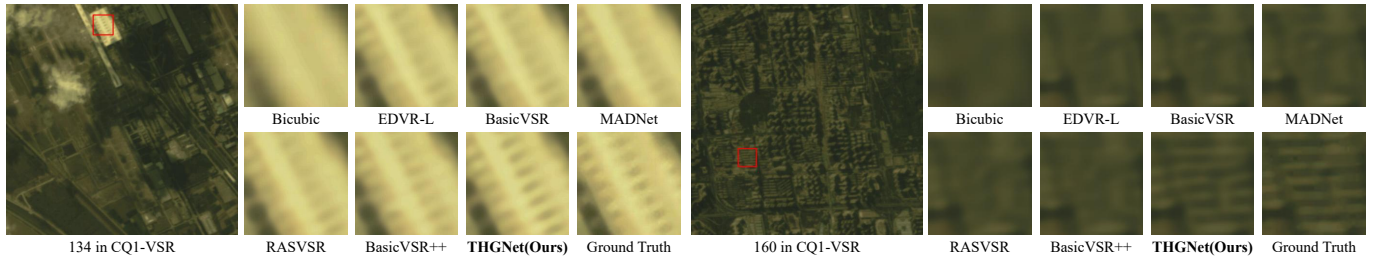


Fig. 10. Qualitative comparison on CQ1-VSR. Zoomed in for better visualization.

MADNet [20], and NSBRNet [46]. Except for NSBRNet, all learning-based methods are retrained on the same dataset splits under the $\times 4$ setting, following their official configurations. Since the implementation of NSBRNet was not publicly available at the time of our experiments, we directly report its published result on SAT-MAT-VSR.

1) Quantitative comparison: Table I reports the quantitative results on SAT-MAT-VSR and CQ1-VSR. Under the reported configurations, THGNet obtains the highest PSNR, SSIM, and the lowest NIQE among the compared methods on SAT-MAT-VSR. Specifically, THGNet reaches 39.6817 dB PSNR and 0.9681 SSIM with 8.2M parameters. Compared with EDVR-L, THGNet improves PSNR by 1.8245 dB while using fewer parameters. Compared with recurrent methods, THGNet exceeds BasicVSR, IconVSR, and BasicVSR++ by 1.6305 dB, 1.6251 dB, and 0.5949 dB in PSNR, respectively. Since THGNet keeps the BasicVSR++ propagation pipeline while modi-

fying feature representation and offset estimation, the gain over BasicVSR++ supports the usefulness of the proposed changes under this training protocol.

THGNet also performs favorably against transformer-based and remote-sensing-specific methods under the reported settings. THGNet surpasses VRT by 0.9617 dB and MIA-VSR by 2.1958 dB in PSNR while using fewer parameters than both models. Compared with recent RSVSR methods, THGNet improves PSNR over RASVSR and MADNet by 0.3173 dB and 1.2666 dB, respectively. These results associate the complementary global-context and local-refinement design with improved reconstruction without relying on a substantially larger parameter count.

On CQ1-VSR, THGNet obtains the highest average PSNR and SSIM among the compared methods, reaching 41.4010 dB and 0.9625, respectively. It exceeds BasicVSR++ by 0.1118 dB in PSNR and also obtains the lowest NIQE in the table. CQ1-VSR retains stronger raw-

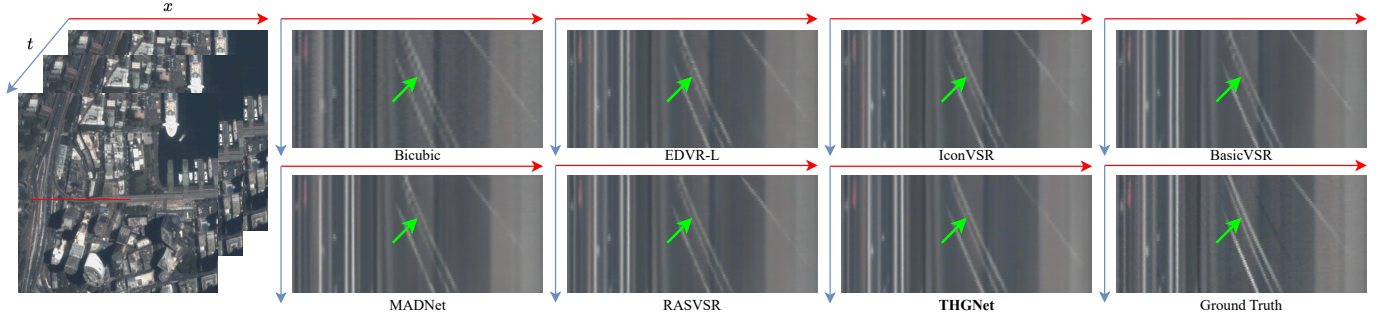


Fig. 11. Qualitative comparison of temporal profiles. We record a row (red line) and stack it across the timeline to observe the temporal changes. The green arrows indicate the two slanted stripe patterns on the right, where THGNet recovers clearer and more continuous temporal trajectories. The dense vertical stripes on the left further show that THGNet preserves sharper static structures and exhibits more consistent temporal profiles.

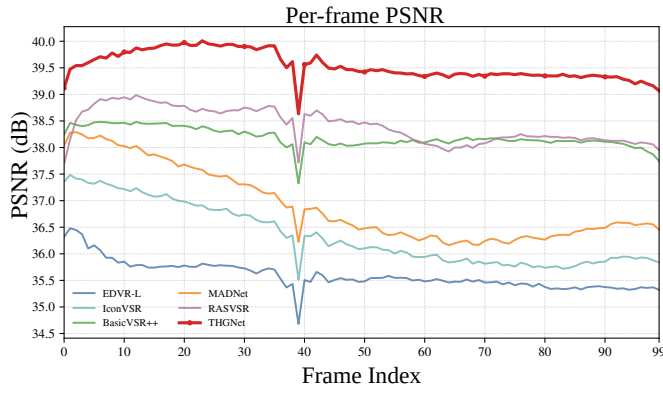


Fig. 12. Quantitative comparison across frame index. The results are conducted on SAT-MAT-VSR val set. Our THGNet achieves consistently superior performance on the entire frame.

source inter-frame displacement and local variation while using synthetically generated LR inputs, so the result indicates that the proposed design remains effective in this evaluated setting. In terms of computational cost, THGNet uses 8.2M parameters and 982.6G FLOPs. It introduces additional cost over BasicVSR++, mainly from HFE and GLOE, although its parameter count remains lower than those of several transformer-based baselines.

2) Qualitative comparison: Fig. 9 and Fig. 10 provide visual comparisons on SAT-MAT-VSR and CQ1-VSR, respectively. In the upper example of Fig. 9, THGNet reconstructs the head of the airplane with a clearer and more recognizable contour, whereas several compared methods produce a blurred front structure. In the lower example, THGNet recovers more complete vehicle bodies and preserves clearer gaps between adjacent vehicles, reducing the visual merging observed in other results.

In Fig. 10, THGNet reconstructs the houses with clearer outlines and sharper local structures, resulting in visibly higher reconstruction quality in the selected regions. This observation is consistent with the motivation for improving offset estimation under stronger inter-frame displacement and local variation, while broader robustness should be

TABLE II
Ablation of Individual Components on SAT-MAT-VSR.

Variant	T-MAE	HFE	GLOE	PSNR (dB)↑	SSIM↑
Variation-A	—	—	—	39.0868	0.9628
Variation-B	✓	—	—	39.3616	0.9656
Variation-C	✓	✓	—	39.4565	0.9663
THGNet	✓	✓	✓	39.6817	0.9681

TABLE III
Ablation of Individual Components on CQ1-VSR.

Variant	T-MAE	HFE	GLOE	PSNR (dB)↑	SSIM↑
Variation-A	—	—	—	41.2892	0.9610
Variation-B	✓	—	—	41.3064	0.9613
Variation-C	✓	✓	—	41.3257	0.9617
THGNet	✓	✓	✓	41.4010	0.9625

interpreted together with the average quantitative results.

These qualitative differences are consistent with the intended roles of the proposed components. GLOE combines global-context recalibration with multi-scale local refinement for offset prediction. T-MAE provides a context-sensitive initialization through temporal feature masking, while HFE supplies high-frequency residual cues before propagation. Together, these components improve temporal feature aggregation and local structural reconstruction in the selected examples.

To further evaluate temporal coherence and reconstruction consistency over long sequences, we record one row of pixels from each frame and stack it along the temporal axis. As shown in Fig. 11, THGNet produces clearer and more continuous trajectories for the selected moving structures while preserving sharper static patterns, demonstrating more accurate spatial-temporal aggregation. We further evaluate the reconstruction performance across the entire frame sequence in Fig. 12. THGNet consistently achieves the highest PSNR across the evaluated frames, showing its ability to aggregate spatial-temporal information for accurate and consistent reconstruction.

TABLE IV
Average Inference Speed Comparison on 100-Frame Sequences
With LR Size 160×160 .

Method	Time / Seq. (s)	FPS	FPS Drop
BasicVSR++	1.3226	75.61	0.00%
Variation-B	1.6303	61.34	18.88%
Variation-C	1.6480	60.68	19.74%
THGNet	1.8352	54.49	27.93%

TABLE V
Ablation of GLOE Local-Branch Dilation Rates on
SAT-MAT-VSR.

Dilation Rates	PSNR (dB)↑	SSIM↑
(1, 1, 1)	39.5548	0.9674
(1, 2, 3)	39.6817	0.9681
(1, 2, 4)	39.5881	0.9671
(1, 3, 5)	39.5693	0.9672

D. Ablation Studies

We conduct ablation studies to analyze the contribution of each component. We first examine the overall component configuration and then analyze the alignment-centered GLOE module, T-MAE pretraining, and HFE in this order.

1) Effectiveness of Main Components: Tables II and III report the contribution of the three proposed components. Variation-A denotes the BasicVSR++ baseline, Variation-B adds T-MAE, Variation-C further adds HFE, and THGNet introduces GLOE on top of Variation-C. The detailed analyses below follow the order of GLOE, T-MAE, and HFE to remain consistent with the method presentation.

On SAT-MAT-VSR, introducing GLOE on top of Variation-C improves PSNR from 39.4565 dB to 39.6817 dB, yielding the largest incremental gain along the reported cumulative configuration. T-MAE improves the BasicVSR++ baseline from 39.0868 dB to 39.3616 dB, supporting the usefulness of temporal feature-masking pretraining under this protocol. HFE further increases the result to 39.4565 dB by supplementing explicit high-frequency cues. On CQ1-VSR, adding GLOE to Variation-C again produces the largest incremental gain in the cumulative study. Because the table evaluates one addition order rather than all module combinations, these gains should not be interpreted as a complete factorial attribution of the three components. Table IV further reports the inference speed of the main component variants, providing an efficiency comparison alongside the reconstruction accuracy.

2) Dilation Rates in GLOE: To analyze the local branch of GLOE, we evaluate several dilation-rate combinations. The setting (1, 1, 1) removes multi-scale dilation, (1, 2, 3) is the default continuous configuration, (1, 2, 4) enlarges the receptive field in a pyramid manner, and (1, 3, 5) tests a sparser large-receptive-field setting.

Since all dilation settings use the same convolution kernels and channels, changing the dilation rates does not increase the parameter count. This experiment therefore

TABLE VI
Ablation of GLOE Local-Branch Dilation Rates on CQ1-VSR.

Dilation Rates	PSNR (dB)↑	SSIM↑
(1, 1, 1)	41.2524	0.9616
(1, 2, 3)	41.4010	0.9625
(1, 2, 4)	41.3988	0.9622
(1, 3, 5)	41.3954	0.9623

TABLE VII
Ablation of Global and Local Branches in GLOE on
SAT-MAT-VSR.

Variant	Global	Local	PSNR (dB)↑	SSIM↑
Variation-C	–	–	39.4565	0.9663
Variation-C+Global	✓	–	39.5321	0.9675
Variation-C+Local	–	✓	39.5040	0.9670
THGNet	✓	✓	39.6817	0.9681

examines how the sampling pattern of the local branch affects offset prediction. If overly sparse dilation weakens performance, it suggests that RSVSR alignment benefits more from continuous local-to-medium receptive fields than from large sparse sampling.

As shown in Tables V and VI, the default (1, 2, 3) setting achieves the best PSNR and SSIM on both datasets. The (1, 1, 1) setting removes multi-scale sampling and gives clearly lower results, showing that a single local receptive field is insufficient for offset estimation. Larger or sparser settings, such as (1, 2, 4) and (1, 3, 5), produce competitive but slightly inferior performance. This suggests that continuous dilation provides a better balance between local offset refinement and medium-range motion modeling in GLOE.

3) Global and Local Branches in GLOE: We further analyze the global and local branches of GLOE based on Variation-C. The global branch is designed to emphasize scene-level alignment evidence through lightweight attention, while the local branch provides multi-scale spatially varying cues.

As shown in Table VII, introducing either branch improves the Variation-C baseline, indicating that both processing paths are useful for second-order deformable alignment. Using only the global branch improves PSNR from 39.4565 dB to 39.5321 dB and SSIM from 0.9663 to 0.9675, which is consistent with the intended benefit of scene-level context recalibration. Using only the local branch also brings a gain, increasing PSNR to 39.5040 dB and SSIM to 0.9670, suggesting that multi-dilation features provide useful cues for spatially varying offset refinement. When the two branches are combined, THGNet reaches 39.6817 dB PSNR and 0.9681 SSIM, outperforming Variation-C by 0.2252 dB and 0.0018 SSIM. The larger combined gain suggests complementary effects under the reported configuration, although the experiment does not impose or verify a unique physical decomposition of the predicted motion.

4) T-MAE Pretraining and Encoder Updating: To examine whether the improvement is associated with

TABLE VIII
Ablation of T-MAE Pretraining and Encoder Updating Strategy on SAT-MAT-VSR. All Settings Use the Same Downstream THGNet Architecture.

Encoder Source	Encoder Update	PSNR (dB) $\uparrow$	SSIM $\uparrow$
Random	Fine-tuned	38.6715	0.9588
T-MAE	Frozen	39.2951	0.9659
T-MAE	Fine-tuned	39.6817	0.9681



Fig. 13. Qualitative evaluation of the structural completion capability of T-MAE. In the selected examples, T-MAE reconstructs more complete object shapes and produces more plausible structures in some local regions.

temporal feature-masking pretraining rather than merely with a deeper feature extractor, we compare settings with the same encoder architecture, i.e., 64 channels and 30 residual blocks. The random-initialization baseline trains this encoder from scratch together with the downstream THGNet, while the T-MAE settings initialize the encoder with the proposed masked pretraining. We further compare frozen and fine-tuned T-MAE encoders to examine whether the pretrained representation should remain fixed or be adapted during downstream VSR training.

As shown in Table VIII, random initialization obtains 38.6715 dB PSNR and 0.9588 SSIM, which is lower than both T-MAE settings under the same encoder architecture. This result indicates that the observed improvement is associated with temporal feature-masking pretraining rather than encoder depth alone. Freezing the T-MAE encoder improves the result to 39.2951 dB and 0.9659 SSIM, suggesting that the pretrained representation transfers useful contextual information to RSVSR. However, keeping the encoder fixed is still suboptimal. When the pretrained encoder is jointly fine-tuned with the downstream VSR network, the performance further increases to 39.6817 dB and 0.9681 SSIM. Therefore, we adopt the fine-tuned T-MAE encoder as the default setting in THGNet.

Fig. 13 evaluates the reconstruction behavior of T-MAE itself. In the selected examples, T-MAE recovers more complete object contours and produces more plausible structures in some local regions, particularly where the available observations provide weak or ambiguous structural evidence. This result provides qualitative evidence that temporal masked reconstruction encourages T-MAE to infer missing structural content from the available

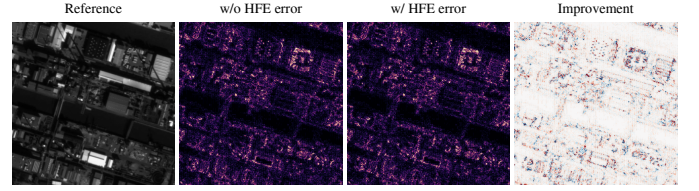


Fig. 14. High-frequency response analysis of HFE. The response maps visualize local high-frequency details in the reconstructed results.

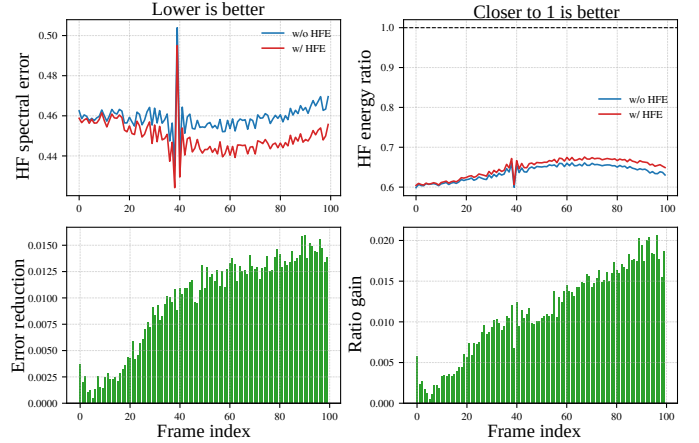


Fig. 15. Radial frequency-profile comparison. The profiles compare the frequency distributions of the results without HFE and with HFE against the GT.

spatial-temporal context.

5) High-Frequency Extraction Strategy: We isolate the effect of the high-frequency operator by using Variation-B as the baseline and enabling HFE while keeping GLOE disabled. The overall HFE structure is kept unchanged; only the extraction operator is replaced.

The Gaussian variant corresponds to the default HFE design. The Laplacian and Sobel variants use standard fixed edge operators, while the learned variant uses a depth-wise learnable filter initialized from a Laplacian kernel. Since the learned filter introduces only 27 additional parameters, its parameter overhead is negligible. This ablation examines whether the observed gain depends on the general high-frequency guidance mechanism or on the specific Gaussian residual design.

As shown in Table IX, the default Gaussian high-pass design achieves the highest PSNR and SSIM among the evaluated variants. Sobel and the learned filter obtain close but slightly lower results, indicating that directional gradients and learnable high-pass filters can also provide useful detail cues. In contrast, the Laplacian operator performs the worst, possibly due to its sensitivity to noise and low-contrast fluctuations. These results support the use of the Gaussian residual as the default high-frequency extraction strategy for HFE under this configuration.

To further analyze the effect of adding HFE to the framework, Fig. 14 visualizes the high-frequency response

TABLE IX
Ablation of High-Frequency Extraction Operators in HFE on SAT-MAT-VSR.

HFE Variant	Extraction Operator	PSNR (dB)↑	SSIM↑
w/o HFE	-	39.3616	0.9656
Gaussian	$I - G(I)$	39.4565	0.9663
Laplacian	Fixed Laplacian	39.3284	0.9654
Sobel	Gradient magnitude	39.4432	0.9661
Learned	Learnable depth-wise filter	39.4296	0.9660

maps. The results show that HFE enhances local high-frequency regions related to edges and fine structures.

Fig. 15 presents the radial frequency profiles. The result with HFE has a frequency distribution closer to the GT than the result without HFE, especially in the high-frequency range. Together with the quantitative gain in Table IX, these observations are consistent with the intended role of HFE in preserving high-frequency information.

V. Conclusion

In this paper, we propose THGNet, a recurrent network developed for remote sensing video super-resolution. The work is motivated by the alignment difficulty caused by large inter-frame displacement and local structural variations in satellite videos. The key design of THGNet is GLOE, which improves second-order deformable alignment through complementary global-context recalibration and multi-scale local refinement. To support this alignment-centered framework, we further introduce T-MAE and HFE. T-MAE provides a context-sensitive initialization for the encoder, while HFE injects gated high-pass residual cues before temporal propagation. We also construct CQ1-VSR from raw unregistered satellite source videos to complement SAT-MAT-VSR with stronger inter-frame variation. Under the reported configurations, THGNet achieves favorable performance against representative recurrent, transformer-based, and remote sensing VSR methods.

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